

Stony Brook University
Alejandro Melo Ponce

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ASSIGNMENT: MIDTERM I PREPARATION

Due: Optional, but you should use it to prepare for the midterm.

INSTRUCTIONS: This is an optional assignment whose purpose is to prepare you for the midterm. It consists of six problems. I strongly recommend that you attempt to prepare all questions. On Tuesday's Midterm I will pick four questions at random from this assignment which you will need to answer.

1. Michele, who has a relatively high income I , has altruistic feelings toward Sofia, who lives in such poverty that she essentially has no income. Suppose Michele's preferences are represented by the utility function

$$U_M(c_M, c_S) = c_M^{1-\alpha} c_S^\alpha,$$

where c_M and c_S are Michele and Sofia's consumption levels, appearing as goods in a standard Cobb-Douglas utility function. Assume that Michele can spend her income either on her own or Sofia's consumption (through charitable donations) and that \$1 buys a unit of consumption for either (thus, the "prices" of consumption are $p_M = p_S = 1$).

- (a) Argue that the exponent α can be taken as a measure of the degree of Michele's altruism by providing an interpretation of extreme values of $\alpha = 0$ and $\alpha = 1$. What value would make her a perfect altruist (regarding others the same as oneself)?
 - (b) Solve for Michele's optimal choices and demonstrate how they change with α .
 - (c) Suppose that there is an income tax at rate τ , i.e. net income now is just $(1 - \tau)I$. Solve for Michele's optimal choices under the income tax rate.
 - (d) Now suppose that besides the income tax rate τ , there are charitable deductions, so that income spent on charitable deductions is not taxed. Argue that this amounts to changing the price p_S from \$1 to $\$(1 - \tau)$. Solve for the optimal choices under both the income tax rate and charitable deductions. Does the charitable deduction have a bigger incentive effect on more or less altruistic people?
2. Suppose that a fast-food junkie derives utility from three goods—soft drinks (x), hamburgers (y), and ice cream sundaes (z)—according to the utility function

$$U(x, y, z) = x^{0.5} y^{0.5} (1 + z)^{0.5}.$$

Suppose also that the prices for these goods are given by $p_x = 1$, $p_y = 4$, and $p_z = 8$ and that this consumer's income is given by $I = 8$.

- (a) Show that, for $z = 0$, maximization of utility results in the same optimal choices as in the case of a Cobb-Douglas utility function $U(x, y) = x^{0.5} y^{0.5}$. Show also that any choice that results in $z > 0$ (even for a fractional z) reduces utility from this optimum.
- (b) How do you explain the fact that $z = 0$ is optimal here?
- (c) How high would this individual's income have to be for any z to be purchased?

3. Consider the utility function

$$u(x, y) = (x + 2)(y + 3), \quad x \geq 0, y \geq 0$$

with the accompanying budget constraint:

$$p_x x + p_y y \leq I, \quad p_x, p_y, I > 0.$$

- Fix a given utility level $U > 0$ and find an explicit expression for the indifference curve defined by the utility level $U > 0$. Then, derive an explicit expression for the marginal rate of substitution between good x and good y .
- Draw the indifference curve (for this associated level of utility U) and carefully label the graph and its elements.
- Show that the utility function is strictly increasing (and hence monotone) in x and y .
- Now formally state the utility maximization problem and briefly describe its content, in particular what constitutes the choice variables, and what constitutes parameters.
- Provide an argument why in the present utility maximization problem, we can restrict attention to the case where the budget constraint holds as an equality. (The argument should not involve the explicit computation of the optimal choices.)

4. Consider the utility function

$$U(x, y) = \min(2x + y, x + 2y)$$

- Draw the indifference curve for $U(x, y) = 20$. Shade the area where $U(x, y) \geq 20$.
- For what values of $\frac{p_x}{p_y}$ will the unique optimum be $x = 0$?
- For what values of $\frac{p_x}{p_y}$ will the unique optimum be $y = 0$?
- If neither x and y is equal to zero, and the optimum is unique, what must be the value of $\frac{x}{y}$?

5. Consider a consumer with a utility function $U(x, y) = e^{(\ln(x)+y)^{1/3}}$.

- What properties about utility functions will make this problem easier to solve?
- Which of the non-negativity constraints will bind for small I ?
- Derive the Marshallian demand functions and the indirect utility function (using the original utility function).
- Derive the expenditure function in terms of the original utility u .

6. There are two goods, food and clothing, whose quantities are denoted by x and y and prices p_x and p_y respectively. There is a consumer whose income is denoted by I and utility by U . His utility function is

$$U(x, y) = \sqrt{xy}.$$

- Find this consumer Marshallian demand functions. Find the indirect utility function and the expenditure function.
- Initially $I = 100$, $p_x = 1$ and $p_y = 1$. What quantities does the consumer buy, and what is his resulting utility?
- Now the price of food rises to $p_x = 1.21$, while income and the price of clothing are as before. What quantities does the consumer buy and what is his resulting utility?

- (d) Suppose the increase in the price of food was caused by the government levying a tax of 21% on food. What is the government revenue from this tax? *Hint: At the new prices, calculate the optimal consumption bundles (x^*, y^*) . Then calculate $(1.21 - 1)x^*$.*
- (e) If the government wants to compensate the consumer by giving him some extra income, how much extra income would be needed to restore him to the old utility level. (*Hint: Use the expenditure function.*) Is the government's revenue from the tax on the good itself sufficient to provide this compensation? What is the economic intuition of your answer?
- (f) If the government tries to compensate the consumer by giving him enough extra income to enable him to purchase the same quantities as he did at the original income and prices of part (b), how much extra income would the government have to give him? With this income and the new prices, what quantities will the consumer actually buy? What will be his resulting utility?